# Digital Twin Version 2.0 Brainstorming

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## 1. Introduction

## 2. What Already Exists in Version 1.0

1. Backengine processing: orbit and twiss

2. Mapping from ”device” view to ”design” view

3.

4.

5.

# 3. Ideas and Opinions for Version 2.0

**Turn by Turn data** or *Tracking*

**Idea** provide turn by turn data of the beam: need to handle input parameters: turns to track, start vector, start position or element, where to output the data to

**Opinion**

**Effort**

**Status** first implementation exists: provides turn by turn data for BPMS next to fall off value

**Pinhole Camera Idea** generate a 2D image of the beam from the beam size at a given location

**Beam current measurement Idea** test bench for systems that need beam current (where does the beam be stopped ?) fixed value in [A] or $\approx 0$ if beam is lost

**Question** additional variable if beam is lost by calculator? Status of machine is enough

**inputs validated by hardware limits idea** Twin settings shall be validated that these are within the limits of the available hardware

**implementation** Any value received or produced in the *design* space has to be converted to the device space. There it needs to be checked with the limits provided by the available hardware. **Todo:** Only "static" limits: for power converters: maximum / minimum current

**Different lattices Idea** load different lattices: be it slightly different lattice or lattice with error or a lattice with error and corrections

**Implementation** as long as the names in the lattice are known to liaison manager and translator service changing the lattice should be transparent. User / client space is responsible for taking a snapshot of all settings as well as replaying it to twin after the lattice is set. No effort is made to make this reset smooth (i.e. don't apply it with Hardware in the Loop unless you know this will not create any damage). Investigate the impact of corrections.

**Output with noise Idea** produce output which is more realistic to real world devices: **Todo:** what is it used for e.g. beam reading, pinhole camera, beam current readings

**Improving user experience Idea** review names of devices or methods if names can be improved to be easier to understand: e.g. ring simulator: *reset* $\rightarrow$ *reinject_beam*, *reinit* $\rightarrow$ *reset_lattice_to_nominal*

**From a machine to a machine complex** **Idea** be able to track through the transfer line into the storage ring

**Discuss** variations of kicker strength? Start distribution

# 4. Outlook to requirements for Version 3.0

Please put items here which are not known to be needed further down the road. Please address in particular points that need to be described in details before one starts implementing (most probably these will be identified during discussion of the partners).

## 4.1. Preparing for hardware in the loop

### 4.1.1. User story: including power converter hardware

As understood up to now: real hardware will be operated as tango device e.g. a power converter (on the test station with a magnet ?).

Idea: provide a twin that includes hardware behaviour which is a bit off from ideal. Here provide response function behaviour of magnet power / converter system.

# A. Reference document: next steps for dt4acc

Minutes notes exchanged at from 26. July 2026

## A.1. DT4ACC steps to 1.0

- accept PR update of Tango
- mark current status as branch 'prod/main'
  - at dt4acc
  - at dt4acc-lib
- publish both as version 0.9

## A.2. CI integration

EPICS digital twin for pyaml CI

## A.3. parallel developments at dt4acc

- tree for stabilising development 'next/main': next

### A.3.1. twin for ALS

main target: turn-by-turn output for the BPM's
data model for ALS machine (seems to be derived from LOCO) for different energies
get its own repo

### A.3.2. add a todo list

- review how to address reference energy / current machine state energy <- booster
- review TANGO: process separation (discussed with you)
- review used multipoles: align what is used in community

## A.4. Soleil further developments

- power converter
- launching devices: input from Guillaume (sep)

### A.4.1. Test cases: m-to-n connection

Sextupole magnets: cross talk with corrector talks (h/v steerers)
current -> multipole multipole <-current
use neural networks for handling m-to-n
Address from element to element at the same time?

### A.4.2. What happens at BESSYII

- user interface current. (device view / engineering)
- lattice (design view / physics)
- command execution engine: rewrites input commands from device view (engineering) to lattice view (physics)